

Chalo shuru karte hain one of the most interactive chapters jo hum 9th class se padhte aa rahe hain... waise toh is chapter ka asli mazaa 12th ke Electrostatics ke baad aata hai because (bcz) Gravitation aur Electrostatics mein kaafi similarity hai... Gravitation Mains ke POV (Point of View) se bohot important hai jahaan few selected articles se repeatedly questions (ques) pooche jaa rahe hain. Woh selected articles main is chapter ke last mein bataaunga... Advance mein is chapter se kaafi kam questions pooche gaye hain, toh chalo shuru karte hain.



NEWTON'S LAW OF GRAVITATION

- ★ The magnitude Force of attraction between two point mass m_1 and m_2 separating at a distance of r is proportional to product of mass m_1 and m_2 and inversely proportional to r^2 .

$$F \propto \frac{m_1 m_2}{r^2}$$

$$F = \frac{G m_1 m_2}{r^2}$$

$$G = 6.67 \times 10^{-11} \frac{\text{Nm}^2}{\text{kg}}$$

Universal Gravitational constant
(Bhai ise yaad krlena kabhi-kabhi exam time me hum yhi bhul jate hai)

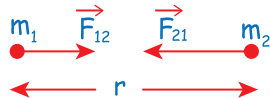
$$\star |\vec{F}_{12}| = |\vec{F}_{21}| = F$$

- ★ Always attractive

- ★ direction of $\vec{F} \rightarrow$ always along the line joining two masses

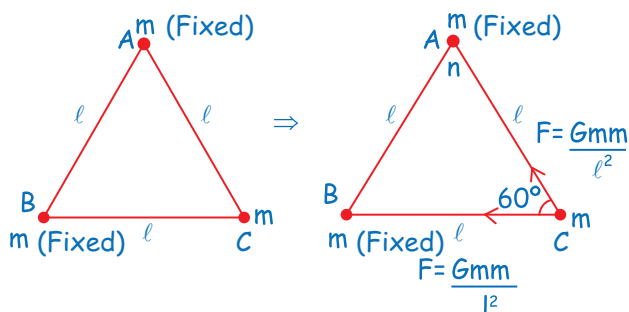
- ★ Gravitational force is an example of Conservative force

- ★ Gravitational force between two particles form an action reaction pair.



- ★ If there are more than two point objects then the net force on any particular object is equal to the vector sum of forces due to all other objects (Super position).

Q. Find Net Gravitational force on mass placed at point 'c'?

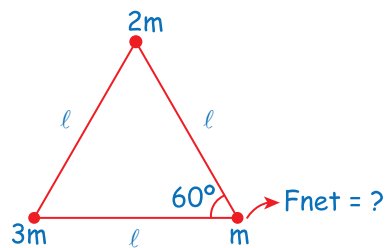


Sol. Net gravitational force on mass at 'c'.

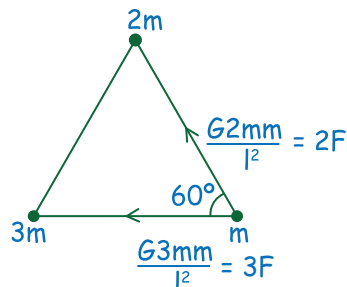
$$|\vec{F}_{\text{net}}| = \sqrt{(F)^2 + (F)^2 + 2(F)(F)\cos 60^\circ}$$

$$= F\sqrt{3} = \frac{Gm^2}{l^2} \sqrt{3}$$

Q. Find F_{net} on mass 'm'?



Sol.

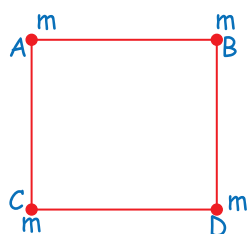


$$\text{let } \frac{Gmm}{l^2} = F$$

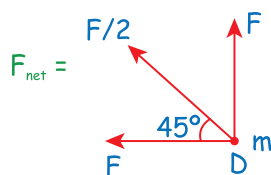
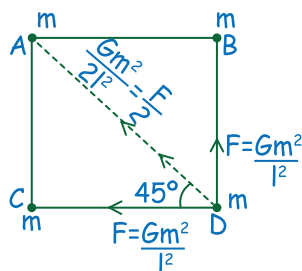
$$F_{\text{net}} = \sqrt{(2F)^2 + (3F)^2 + 2 \times 2F \times 3F \times \cos 60^\circ}$$

$$F_{\text{net}} = F\sqrt{19} = \frac{Gm^2}{l^2} \sqrt{19}$$

Q. Find F_{net} on mass at point D.

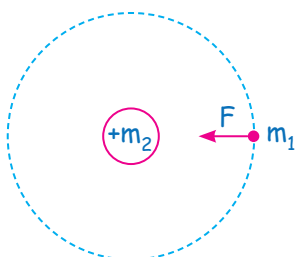


Sol.



$$F_{\text{net}} = F\sqrt{2} + \frac{F}{2} = F\left(\sqrt{2} + \frac{1}{2}\right) = \frac{Gm^2}{l^2}\left(\sqrt{2} + \frac{1}{2}\right)$$

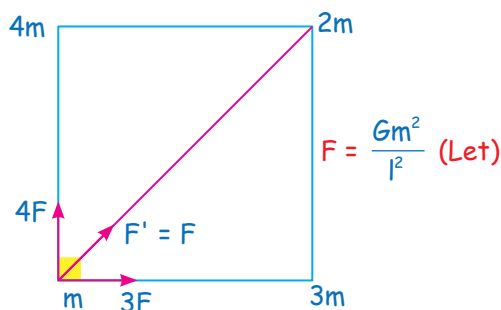
Q. A mass m_1 is moving in a circular path of radius r around a mass m_2 with constant speed. Find speed.



Aisa toh Bohr's Model ke first postulate mein bhi hota tha aur satellite motion mein bhi yahi hota hai.

Sol. $F = \frac{Gm_1m_2}{r^2} = \frac{m_1v^2}{r} = m_1r\omega^2$

Q. Find net gravitational force on point mass m.



$$F = \frac{Gm^2}{l^2} \text{ (Let)}$$

Teenon Force ka resultant will be the answer. 3F aur 4F ka resultant 5F aayega. So, agar tumne answer = 5F + F = 6F diya, toh tum bohot bade gadhe (idiots/fools) ho.

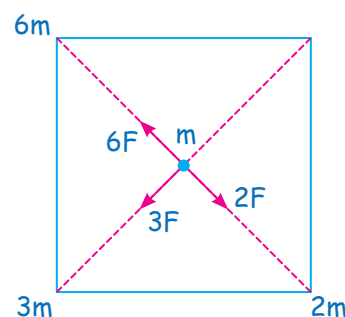
Because (Bcz) 3F aur 4F ka resultant 5F toh aayega, but woh along the F nahi hoga (not along the direction of F), so, we have to solve vectorly.



$$F' = \frac{2Gm^2}{(l\sqrt{2})^2} = F$$

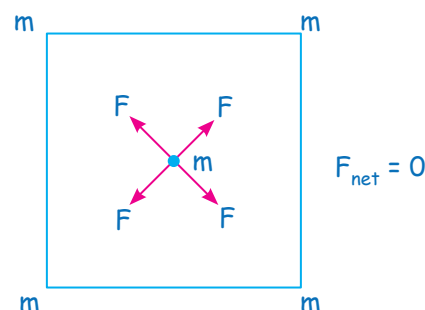
$$F_{\text{net}} = 3F\hat{i} + 4F\hat{j} + \left(+\frac{F\hat{i}}{\sqrt{2}} + \frac{F\hat{j}}{\sqrt{2}} \right)$$

Q. Find net gravitational force on point mass m.



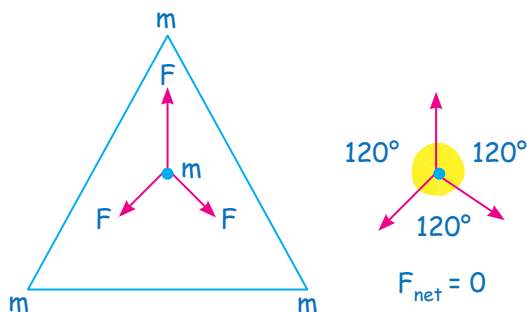
$$F_{\text{net}} = 5F = \frac{5Gm^2}{(l/\sqrt{2})^2} = \frac{10Gm^2}{l^2}$$

Q. Find net gravitational force on point mass m and centre of square.

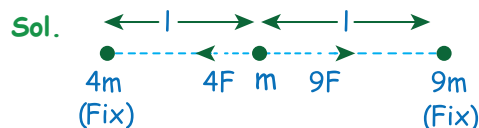
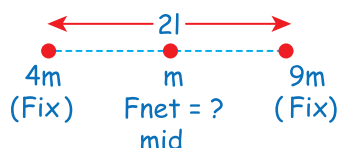


$$F_{\text{net}} = 0$$

Q. Find F_{net} on mass m at centroid of equilateral triangle.



Q. F_{net} on 'm'?

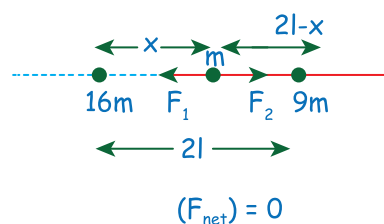


$$F_{\text{net}} = 9F - 4F = 5F = \frac{5Gm^2}{l^2}$$

Q. Where we should place point mass 'm' such that net force on this mass become zero?



Sol. Let the mass 'm' be kept at 'x' dist. from 16m



We must have

$$\frac{G(16m)(m)}{x^2} = \frac{G(9m)m}{(2l-x)^2}$$

$$\frac{4}{x} = \frac{3}{(2l-x)}$$

$$8l - 4x = 3x$$

$$x = \frac{8}{7}l \text{ From 16m}$$

$$2l - x = 2l - \frac{8l}{7} = \frac{6l}{7} \text{ (From 9m)}$$

SKC

Null point dono k bich mein chhote wale mass k karib hoga

Q. Two point masses having separation between them $2r$ are moving in circular path about the common centre of mass in a radius r . Find their speed?



Here Gravitational force acts as Centripetal force.

$$F_c = \frac{mv^2}{r^2} = \frac{Gmm}{(2r)^2} \text{ (Centripetal force)}$$

$$\frac{Gmm}{(2r)^2} = \frac{mv^2}{r}$$

$$v = \sqrt{\frac{Gm}{4r}}$$

Next question is very besharam exam Mein ata rhta h



Q. 4 equal mass 'm' which are kept at the edges of square of side ' ℓ ' are moving in a circular path of radius r as shown in diagram under their mutual force of Attraction find angular speed (ω)?

Sol. $F_{\text{net}} = \frac{F}{2} + F\sqrt{2} = \frac{mv^2}{r}$

$$F\left(\frac{1}{2} + \sqrt{2}\right) = \frac{mv^2}{r}$$

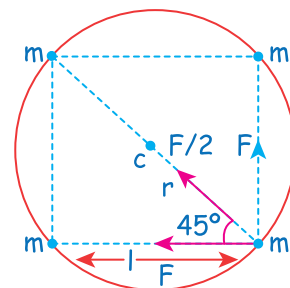
$$\frac{Gmm}{l^2}\left(\frac{1}{2} + \sqrt{2}\right) = \frac{mv^2}{r}$$

$$= mr\omega^2$$

Form fig. we have $r \cos 45^\circ = \frac{\ell}{2}$ or $\left(r = \frac{\ell}{\sqrt{2}}\right)$

$$\text{So, } \omega^2 = \frac{1}{r} \frac{Gm}{\ell^2} \left(\sqrt{2} + \frac{1}{2}\right) \Rightarrow \omega \sqrt{\frac{Gm}{\ell\sqrt{2}}} \left(\sqrt{2} + \frac{1}{2}\right)$$

Where $r \cos 45^\circ = \frac{\ell}{2}$ (from fig.)



Q. Find the force applied by rod (m_2 , l) on point mass m_1 as shown in fig ?

Sol.

$$dF = \frac{Gm_1 dm}{x^2}$$

$$dm = \lambda dx \quad (\text{mass of small element on Rod})$$

$$= \frac{m_2}{l} dx$$

$$dF = \frac{Gm_1 m_2}{x^2} dx$$

$$\int dF = \int_a^{a+l} \frac{Gm_1 m_2}{l x^2} dx$$

$$F = \frac{Gm_1 m_2}{l} \left[-\frac{1}{x} \right]_a^{a+l}$$

$$F = \frac{Gm_1 m_2}{a(a+l)}$$

If $a \gg l$ (Rod kept far away)

$$F = \frac{Gm_1 m_2}{a(a+0)} = \frac{Gm_1 m_2}{a^2}$$

GRAVITATION FIELD (E_g)

All the bodies on or above earth's surface experience gravitational force known as the weight of the bodies. So, the space surrounding the earth, where the gravitational force (weight) is experienced is said to contain the gravitational field of the earth. Gravitational field is a vector quantity.

★ It is defined as (E_g) = Gravitation Force per Unit test mass.

$$E_g = \lim_{m_0 \rightarrow 0} \frac{F_{net}}{m_0} = \frac{Gmm_0}{r^2 m_0} = \frac{Gm}{r^2}$$

★ Independent on test mass

Gravitation

SKC

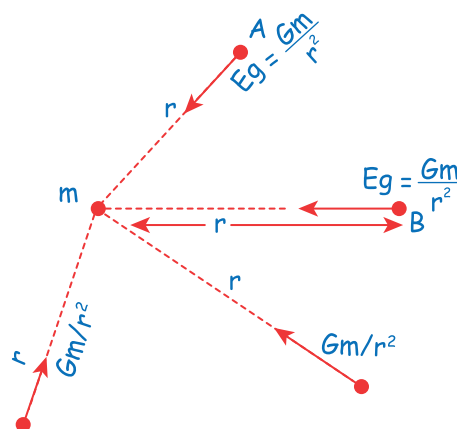
★ Dekh bhai, mudde ki baat yeh hai ki ek point mass m ki wajah se r distance doori par gravitational field ki value Gm/r^2 hogi, jiski direction towards the mass m hogi.

★ Agar main kisi jagah par 1 kg ka mass rakh doon toh us par jitna Gravitation Force lagega, utne hi magnitude ki Gravitational Field (Grav. field) us jagah par hogi. (N/kg Unit SI)



GRAVITATIONAL FIELD DUE TO POINT MASS

★ Grav. Field due to point mass is always directed towards the mass and has value $\frac{Gm}{r^2}$



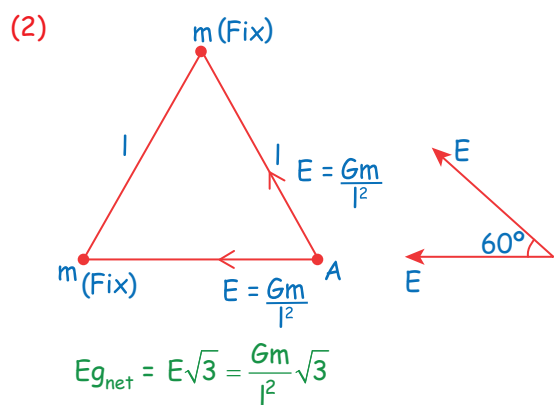
$$\text{Since, } \vec{E}_g = \frac{\vec{F}_{net}}{m} = \vec{F}_{net} = m(\vec{E}_g)$$

Hence we can say that

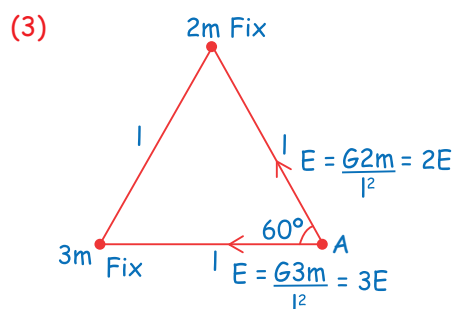
If a point mass (m') is placed at a point where Gravitational field is E_g then it will experience force equal to $m'E_g$.

Q. Find Gravitational field at point A in each cases.

(1) $E_g = \frac{Gm}{r^2}$

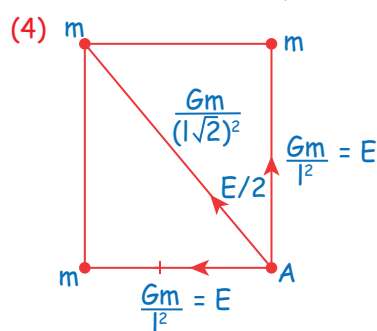


In dono question ko dekh k ab tak tum samajh gye honge ki ye questikn bilkul aise hi h jaise hmne pahle net force nikalna sikha tha bas yha par gravitational field puch liya



$$E_g \text{ net at A} = \sqrt{(2E)^2 + (3E)^2 + 2 \times 2E \times 3E \cos 60^\circ}$$

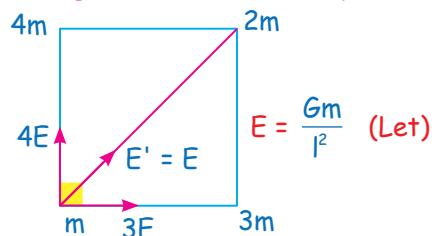
$$= E\sqrt{19} = \frac{Gm}{l^2}\sqrt{19}$$



(Square length, l)

$$E_{g \text{ net}} = \left(E\sqrt{2} + \frac{E}{2} \right) = \frac{Gm}{l^2} \left(\sqrt{2} + \frac{1}{2} \right)$$

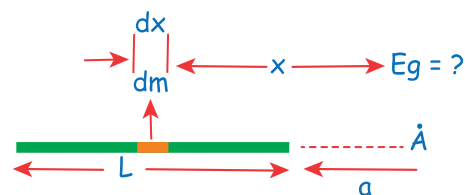
Q. Find net gravitational field on point A.



$$E' = \frac{2Gm}{(l\sqrt{2})^2} = E$$

$$E_{net} = 3E\hat{i} + 4E\hat{j} + \left(+\frac{E\hat{i}}{\sqrt{2}} + \frac{E\hat{j}}{\sqrt{2}} \right)$$

Q. Find grav field at A due to uniform rod of mass m and length L in following figure.



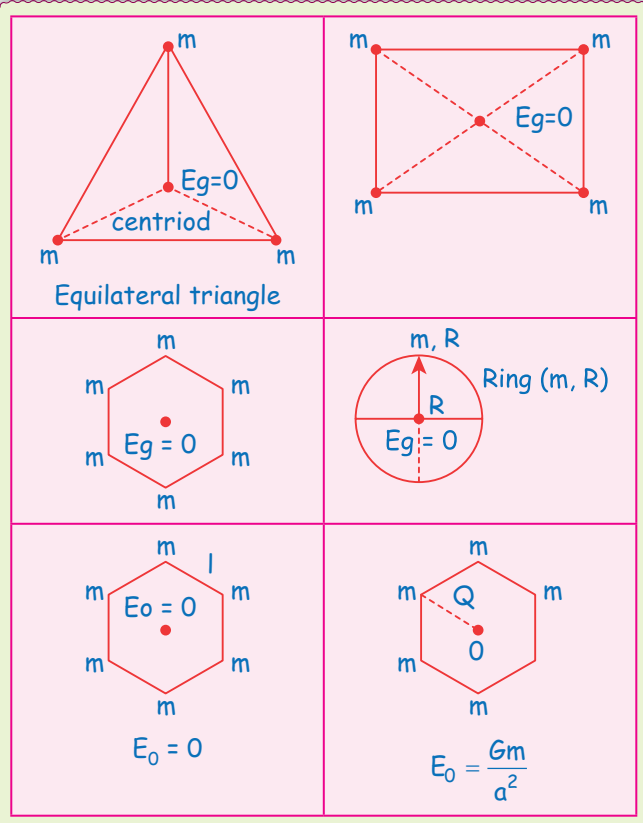
$$dm = \frac{m}{L} dx$$

$$\int dE = \int_a^{a+L} \frac{Gdm}{x^2} = \int_a^{a+L} \frac{G \frac{m}{L} dx}{x^2}$$

$$E_g = \frac{Gm}{a(a+L)}$$

Bhaiya hmsne x pr jakr dx length ka ek chhota sa point mass dm pakra jiske wajah se A point par grav field dF hai

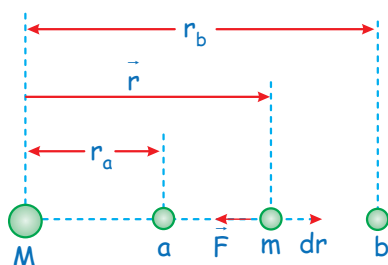
SYMMETRIC CASES



GRAVITATIONAL POTENTIAL ENERGY

Since gravitational force is conservative so we can calculate the potential energy.

Figure shows a particle of mass m moving from a to b along a radial path. A particle of mass M , which we assume to be at rest at the origin, exerts a gravitational force on m . The vector $\vec{r}$ locates the position of m relative to M .



The work done by the gravitational force is

$$W_{ab} = \int_a^b \vec{F} \cdot d\vec{r} = - \int_a^b F dr$$

($\because \vec{F}$ and $d\vec{r}$ are in opposite direction)

$$= - \int_{r_a}^{r_b} \frac{GMm}{r^2} dr = -GMm \int_{r_a}^{r_b} \frac{dr}{r^2}$$

$$= -GMm \left(-\frac{1}{r} \right) \Big|_{r_a}^{r_b} = GMm \left(\frac{1}{r_b} - \frac{1}{r_a} \right)$$

Applying equation ($\Delta U = -W_{ab}$), we can find the change in the potential energy of the system as m moves between points a and b .

$$\Delta U = U_b - U_a = -W_{ab} = GMm \left(\frac{1}{r_a} - \frac{1}{r_b} \right)$$

Instead of differences in potential energy, we can consider the value of the potential energy at a single point if we define a reference point. We choose our reference point to be an infinite separation of the particles, and we define the potential energy to be zero at that point. Let us evaluate equation for $r_b = \infty$ and $U_b = 0$. If a represents any arbitrary point, where the separation between the particles is r , then the above equation becomes

$$U(\infty) - U(r) = GMm \left(\frac{1}{r} - 0 \right)$$

$$\text{or } U(r) = -\frac{GMm}{r}$$

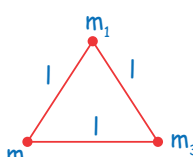
$$\star \Delta U = (WD)_{\text{ext. force}}$$

Dekh bhai mudde ki bat ye h ki agar do point mass m_1 and m_2 , r distance Duri pr h to unki potential energy $U = \frac{-Gm_1m_2}{r}$ Hogi isme minus galti se v mt bhulnq chlo is pr kch question krte h taki ye aur clear ho jaye are you ready....



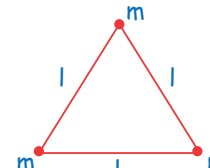
Q. Find P.E of system in following cases

(1) $m_1 \xrightarrow{r} m_2 \quad U = \frac{-Gm_1m_2}{r}$

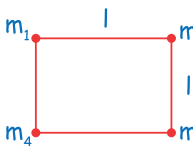
(2) 

$$U = -\frac{Gm_1m_2}{l} + \frac{-Gm_1m_3}{l} + \left(\frac{-Gm_2m_3}{l} \right)$$

$$U = -\frac{Gm_1m_2}{l} - \frac{Gm_1m_3}{l} - \frac{Gm_2m_3}{l}$$

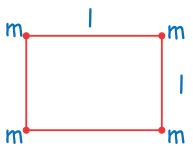
(3) $m \rightarrow \text{same}$


$$U = -\frac{Gmm}{l} \times 3$$

(4) 

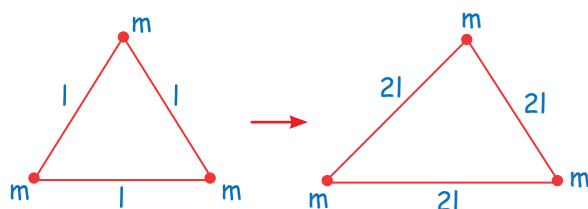
$$U = -\frac{Gm_1m_2}{l} - \frac{Gm_2m_3}{l} - \frac{Gm_3m_4}{l}$$

$$- \frac{Gm_1m_4}{l} - \frac{Gm_1m_3}{l\sqrt{2}} - \frac{Gm_2m_4}{l\sqrt{2}}$$

(5) 

$$U = \left(-\frac{Gmm}{l} \times 4 \right) - \left(\frac{Gmm}{l\sqrt{2}} \right) \times 2$$

Q. Following change in configuration of masses find the (WD)_{ext}?



Sol. $\therefore$ (WD)_{ext} = $U_f - U_i = \Delta U$

$$U_i = \frac{-Gmm}{l} \times 3 \text{ (initial configuration of masses)}$$

$$U_f = \frac{-Gmm}{2l} \times 3 \text{ (final configuration of masses)}$$

Now we know

$$(WD)_{\text{ext agent}} = \Delta U$$

$$(WD)_{\text{ext}} = \frac{-3Gm^2}{2l} - \left(\frac{-3Gm^2}{l} \right) = +\frac{3Gm^2}{2l}$$

SATELLITE MOTION

Suppose a satellite of mass m_2 is moving around the earth (of mass m_1) in an orbit of radius r under the action of gravitational force. Gravitational force provided necessary centripetal force

$$\therefore F = \frac{Gm_1m_2}{r^2} = \frac{m_2v^2}{r}$$

$$v = \sqrt{\frac{Gm_1}{r}} = \sqrt{\frac{Gm_c}{r}} \text{ (Independent of } m_2)$$

orbital velocity radius of orbit

(m_2 : mass of satellite)

Top view

★ Time period

$$T = \frac{2\pi r}{v} = 2\pi r \sqrt{\frac{r}{Gm}}$$

$$\text{or } T = 2\pi \sqrt{\frac{r^3}{Gm}}$$

$$T^2 \propto r^3$$

★ Angular speed.

$$\omega = \frac{2\pi}{T} = \sqrt{\frac{Gm_e}{r^3}}$$

★ Kinetic Energy.

$$K.E = \frac{1}{2} m_1 v_0^2 = \frac{1}{2} m_1 \frac{Gm_2}{r} = \frac{-Gm_1m_2}{2r}$$

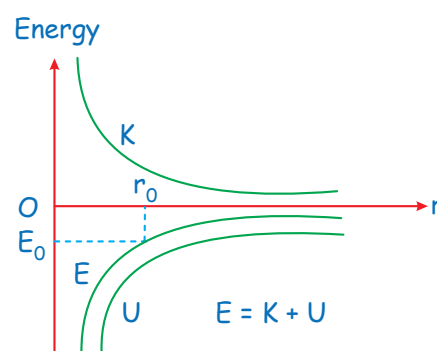
★ Potential Energy.

$$P.E = \frac{-Gm_1m_2}{r}$$

★ Total Energy = KE + PE

$$\text{Total energy} = \frac{Gm_1m_2}{2r} - \frac{Gm_1m_2}{r} = \frac{-Gm_1m_2}{2r}$$

$$|T.E| = K.E = \frac{|P.E|}{2}$$



GEOSTATIONARY SATELLITE MOTION

- ★ They have period of revolution same as earth time period i.e. $T = 24$ hour (approx.)
- ★ $\omega, T \rightarrow$ same as of earth
- ★ Remain at rest wrt earth
- ★ Sense of rotation same as that of earth. It must revolve from west to east.
- ★ It must move in the equatorial plane.
- ★ $h = 3600$ km (approx.) from surface of earth

$$T = 24 \text{ hour} = 2\pi \sqrt{\frac{r^3}{Gm_e}}$$

after putting value of G and m_e we got value of r

$r = 42500$ km (radius of orbit)

Since $r = R + h \Rightarrow h = 36000$ km (approx.)

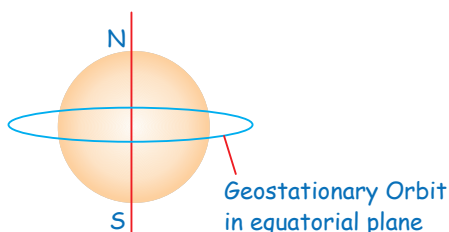
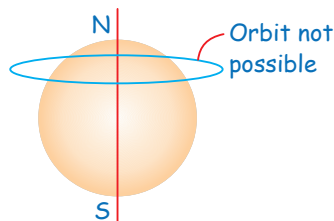
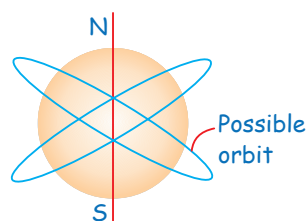
Dekh rha h binod geostationary satellite
Jamin se 36000 km Upar hota h



Q. Find external WD require to increase the orbit of satellite from r_1 to r_2

Sol. $(T.E)_f - (T.E)_i = \text{Ext. WD.}$

$$-\frac{Gm_1m_2}{2r_2} - \left(-\frac{Gm_1m_2}{2r_1}\right)$$



Q. A satellite of mass m is rotated round the earth in a circular orbit of radius $3R$. what extra energy should be given to the satellite if it is to be lifted into orbit of Radius $4R$

Sol. $(T.E)_i = \frac{-Gm_1m_2}{2 \times 3R}$

$$T.E = \frac{Gm_1m_2}{2r}$$

Center se distance

$$-(T.E)_f = \frac{-Gm_1m_2}{2 \times 4R}$$

$$\Delta(T.E) = \left(-\frac{Gm_1m_2}{2 \times 4R}\right) - \left(-\frac{Gm_1m_2}{2 \times 3R}\right)$$

$$= \frac{Gm_1m_2}{2R} \left(\frac{1}{3} - \frac{1}{4}\right) = \frac{Gm_1m_2}{24R}$$

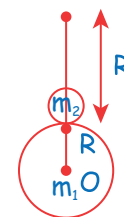
Q. A satellite of mass m initially at rest at earth is to be launched into a circular path at a height of equal to radius of earth. Find the minimum energy required.

Sol. $\rightarrow \text{Energy required} = (T.E)_f - (T.E)_i$

$$(T.E)_i = KE + PE = 0 - \frac{Gm_1m_2}{R}$$

$$(T.E)_f = -\frac{Gm_1m_2}{2 \times (2R)}$$

$$\begin{aligned} \text{Energy required} &= \left(-\frac{Gm_1m_2}{4R}\right) - \left(-\frac{Gm_1m_2}{R}\right) \\ &= \frac{3}{4} \frac{Gm_1m_2}{R} = \frac{3}{4} \frac{Gm_e m_s}{R} \end{aligned}$$



Q. Two satellites s_1 and s_2 of equal masses revolve in the same sense around a heavy planet in coplanar circular orbit of radii R and $4R$ respectively. Find ratio of the time period, velocity, angular momentum, kinetic energy

Sol. $v_1 = \sqrt{\frac{GM}{R}}$ $v_2 = \sqrt{\frac{GM}{4R}} \Rightarrow v_1 = 2v_2$

$$T_1 = \frac{2\pi R}{v_1}, \quad T_2 = \frac{2\pi(4R)}{v_2} \Rightarrow \frac{T_1}{T_2} = \frac{1}{8}$$

$$L_1 = mv_1R, \quad L_2 = mv_24R \Rightarrow \frac{L_1}{L_2} = \frac{1}{2}$$

$$K.E = \frac{Gm_s m_e}{2r} \Rightarrow \frac{K_1}{K_2} = \frac{R_2}{R_1} = \frac{4R}{R}$$

ESCAPE VELOCITY

Kisi particle ko kitni min velocity se feku ki o kbhi laut kr bapas na aye is called escaped velocity bs yhi h



It is the minimum speed with which a body must be projected from the surface of a planet so that particle escapes from gravitational field or particle does not return or particle reaches to infinite.

$$K_i + U_i = K_f + U_f$$

$$\frac{1}{2}mv_e^2 = \frac{Gm_em}{R} = 0 + 0$$

$$V_e = \sqrt{\frac{2Gm_e}{R}} \quad V_e = 11.2 \text{ km/sec}$$

(for earth)

★ Escape speed depends on Mass (M_e) and size (R) of the planet and Position from where the particle is projected.

★ Escape speed does not depend on mass of body and angle of projection.

Q. If escape speed on the surface of earth is v_e , then find the escape speed on the surface of another planet having radius α times the radius of earth and density β times the density of earth.

Sol. $v_e = \sqrt{\frac{2GM}{R}}$ and $M = \rho \frac{4}{3}\pi R^3$

$$\therefore v_e = \sqrt{\frac{2G}{R} \cdot \rho \frac{4}{3}\pi R^3} \Rightarrow v_e = 2R\sqrt{\frac{2G\rho\pi}{3}}$$

$$\text{Now, } \frac{v_{\text{planet}}}{v_{\text{earth}}} = \frac{R_p}{R_e} \sqrt{\frac{\rho_p}{\rho_e}} = \frac{\alpha R_e}{R_e} \sqrt{\frac{\beta \rho_e}{\rho_e}} \Rightarrow \frac{v_p}{v_e} = \alpha\sqrt{\beta}$$

$$\therefore \text{Escape speed on planet} = (\alpha\sqrt{\beta})v_e$$

Q. Escape velocity of a body from earth is 11.2 km/s. If the radius of a planet be one-third the radius of earth and mass be one-sixth that of earth, the escape velocity from the planet is:

[30 Jan, 2024 (Shift-II)]

Sol. Escape velocity of a body from earth = 11.2 km/s.

$$R_p = \frac{R_e}{3}, M_p = \frac{M_e}{6}$$

$$V_e = \sqrt{\frac{2GM_e}{R_e}} \quad \dots(i)$$

$$V_p = \sqrt{\frac{2GM_p}{R_p}} \quad \dots(ii)$$

$$\frac{V_e}{V_p} = \sqrt{2}$$

$$V_p = \frac{V_e}{\sqrt{2}} = \frac{11.2}{\sqrt{2}} = 7.9 \text{ km/sec}$$

The escape velocity from the planet is 7.9 km/sec.

Kepler's Law

- 1 First law (law of orbits): All planets move around the sun in elliptical orbit having the sun at one focus of the ellipse.
- 2 Second law (law of Areas): A line joining any planet to the sun sweep out equal areas in equal time that is the areal speed of the planet remains constant. According to second law. When the planet is nearest to the sun. Then its speed is maximum and when it is farthest from sun, then its speed is minimum.

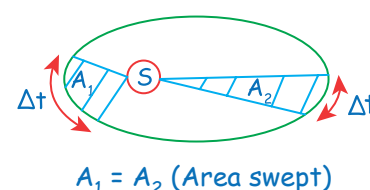
$$A_1 = A_2$$

angular momentum

$$(Z)_{\text{abt sun}} = 0$$

$$Mvr_{\perp} = \text{const}$$

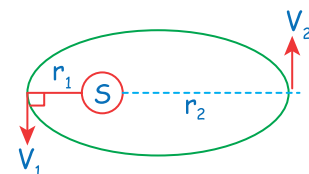
$$m v_1 r_1 = m v_2 r_2$$



$$A_1 = A_2 \text{ (Area swept)}$$

$$\left(\frac{dA}{dt}\right) = \frac{J}{2m} = \text{const}$$

angular momentum
Areal speed



So, The areal speed (dA/dt) of planet is constant
The Second law of Kepler is equivalent to conservation of angular momentum

- 3 Third law (law of periods)

The square of the period of revolution (time one complete revolution) of any planet around the sun is directly proportional to cube of semi-major axis of its elliptical orbit

$$T^2 \propto a^3$$

Ab main niche kuch bodies ki wajah se gravitation al field ka formula likh rha hu ho ske to yad kr lena nhi to koi bat nhi bcz Inka detail derivation and more ques practice hm 12th class Mein electrostatics Me krenge

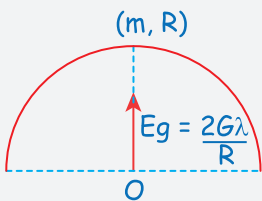
GRAVITATIONAL FIELD DUE TO:

(a) Point Mass



$$E_g = \frac{Gm}{r^2} \text{ (toward mass)}$$

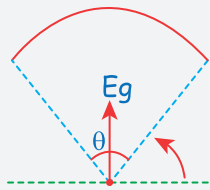
(b) Semi Circular Ring



$$(E_g)_O = \frac{2G\lambda}{R}$$

$$\lambda = \frac{m}{\pi R}$$

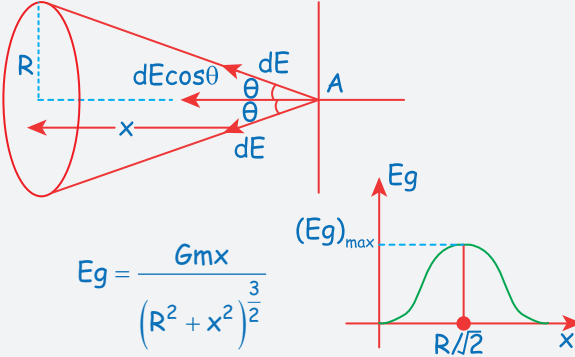
(c) Arc



$$(E_g)_O = \frac{2G\lambda}{r} \sin \frac{\theta}{2}$$

$\lambda = \text{mass per unit length}$

(d) Ring (m, R)



$$E_g = \frac{Gmx}{(R^2 + x^2)^{3/2}}$$

(e) Hollow Sphere

(1) $r < R$, Inside

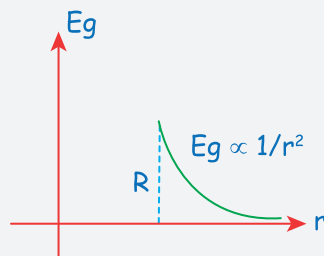
$$E_g = 0$$

(2) $r > R$, at Outside

$$E_g = \frac{Gm}{r^2}$$

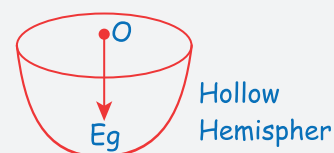
(3) $r = R$, Surface

$$E_g = \frac{Gm}{R^2}$$



(f) Hollow Hemisphere

$$E_g \text{ at } O = \frac{Gm}{2R^2}$$



(g) Solid Sphere (m, R)

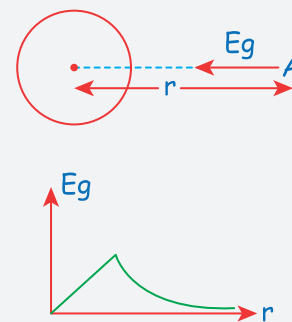
(1) Outside

$$r > R$$

$$(E_g)_A = \frac{Gm}{r^2}$$

(2) Inside ($r < R$)

$$E_g = \frac{Gmr}{R^3}$$



EARTH

Variation of 'g'

- 1 with height

$$g_A = \frac{Gm}{r^2} = \frac{Gm}{(R+h)^2} \quad \text{mass of earth}$$

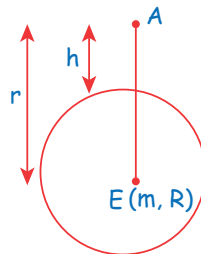
If $h \ll R$

$$g_A = \frac{Gm}{R^2 \left(1 + \frac{h}{R}\right)^2} = \frac{Gm}{R^2} \left(1 + \frac{h}{R}\right)^{-2}$$

$$g_A = \frac{Gm}{R^2} \left(1 - \frac{2h}{R}\right)$$

(Using Binomial Approximation)

$$g_A = g_0 \left(1 - \frac{2h}{R}\right)$$



- 2 At surface

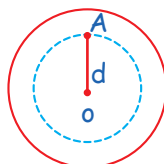
$$g_{\text{surface}} = g_0 = \frac{Gm}{R^2} = 9.8 \text{ m/s}^2$$

$$g_0 = \frac{G\rho \times (4/3)\pi R^3}{R^2} = \frac{4\pi G\rho R}{3}$$

If ' ρ ' is density of earth.

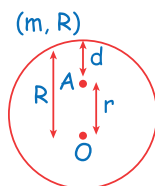
- 3 At depth.

$$g_A = \frac{Gm(R-d)}{R^3} = g_0 \left(1 - \frac{d}{R}\right)$$



- 4 (Eg) at A inside = $g_A = \frac{Gmr}{R^3}$

$$= g_A = \frac{Gm(R-d)}{R^3}$$

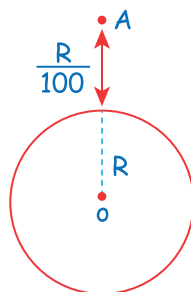


Q. Find the value of g at a height $R/100$ from surface of earth where 'R' is radius of earth.

Sol. $g_A = \frac{Gm}{\left(R + \frac{R}{100}\right)^2}$

$$g_0 = g_0 \left(1 - \frac{2h}{R}\right)$$

$$= g_0 \left(1 - \frac{2 \times R}{100 \times R}\right)$$



$$= g_0 \frac{98}{100} = 0.98 g_0$$

If distance is $\frac{R}{2}$ then

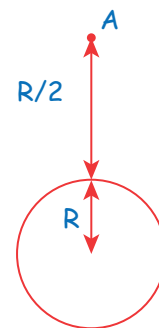
$$g_A = \frac{Gm}{r^2} = \frac{Gm}{\left(\frac{3R}{2}\right)^2}$$

$$g_A = \frac{4}{9} \cdot \frac{Gm}{R^2}$$

$$g_A = \frac{4}{9} g_0$$

g at surface = g_0

$$g_0 = \frac{Gm}{R^2} = \frac{4}{3} \pi G\rho R$$



Q. Planet 1

Planet 2

m

4m

R

3R

g_1

g_2

$$\frac{g_1}{g_2} = ?$$

Sol. $g = \frac{Gm}{R^2}$

$$\frac{g_1}{g_2} = \frac{m_1}{m_2} \times \frac{R_2^2}{R_1^2}$$

$$= \frac{m}{4m} \times \left(\frac{3R}{R}\right)^2 = \frac{9}{4}$$

Q. Find the grav. field inside the sphere at a diameter $\frac{R}{3}$ from centre of sphere of mass m and radius R if

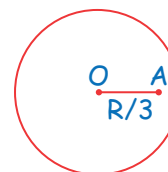
Sol. (1) If Sphere is hollow (shell thin)

$$E_g = 0 \text{ (inside)}$$

(2) If sphere is solid

$$(Eg)_A = \frac{Gmr}{R^3} \text{ (inside)}$$

$$= \frac{Gm \left(\frac{R}{3}\right)}{R^3}$$



Variation of g due to rotation of earth

$$N + mr\omega^2 \cos\theta = mg \dots (i)$$

$$\text{here, } R \cos\theta = r$$

$$N + mr \cos\theta \omega^2 \cos\theta = mg$$

$$N = mg - mR\omega^2 \cos^2\theta$$

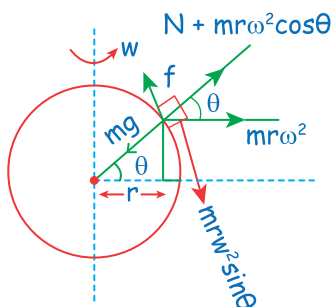
$$N = m(g - R\omega^2 \cos^2\theta)$$

$$g' = g - R\omega^2 \cos^2\theta$$

$$\text{At equator } \theta = 0^\circ; g' = g_0 - R\omega^2$$

$$\text{At Pole } \theta = 90^\circ; g' = g_0$$

As we move from equator to Pole value of g increase (change by $R\omega^2$)



Q. At what rate should the earth rotate so that the apparent g at the equator becomes zero? What will be the length of the day in this situation?

Sol. At earth's equator effective value of gravity is

$$g_{\text{eff}} = g_s - \omega^2 R_e$$

If g_{eff} at equator to be zero,

$$\text{We have } g_s - \omega^2 R_e = 0$$

$$\text{or (Rate of Rotation) } \omega = \sqrt{\frac{g_s}{R_e}}$$

Thus length of the day will be

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{R_e}{g_s}}$$

$$= 2 \times 3.14 \sqrt{\frac{6.4 \times 10^6}{9.8}} = 5075 \text{ s} \approx 84.5 \text{ min.}$$



Kam ka Dabba

$$1 \quad g_0 = \frac{Gm}{R^2} = \frac{4\pi G \rho R}{3}$$

$$2 \quad \text{At height } g = \frac{Gm_e}{r^2} = \frac{Gm_e}{(R+h)^2}$$

$$g = g_0 \left(1 - \frac{2h}{R} \right) \text{ when } h \ll R$$

$$3 \quad \text{At dept } g = g_0 \left(1 - \frac{d}{R} \right)$$

$d \rightarrow$ dept from surface

$$4 \quad g = g_0 - R\omega^2 \cos^2\theta$$

$$5 \quad V_{\text{escape}} = \sqrt{2g_0 R} = \sqrt{\frac{2Gm}{R}}$$

From earth.

$$6 \quad \text{Escape velocity}$$

$$-\frac{Gmm}{R} + \frac{1}{2}mv_0^2 = 0 + 0$$

$$V_0 = \sqrt{\frac{2Gm}{R}}$$

$$V_e = \sqrt{\frac{2Gm}{R} \cdot \frac{R}{R}} = \sqrt{2 \frac{Gm}{R^2} R}$$

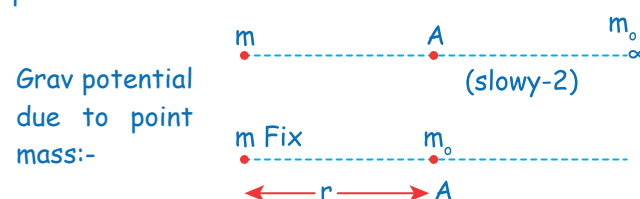
$$v_e = \sqrt{2g_0 R}$$



GRAVITATION POTENTIAL

If a particle is slowly brought without increase in its KE work done by external agent equals to the change in the Potential Energy.

however, 'potential' at a point is defined as work done per unit mass by an external agent. In bringing a particle slowly from reference point to given point. Generally the reference point is chosen as and so that potential at ∞ is zero



$$(wD)_{\text{ext}} = \Delta U = U_f - U_i \quad (\text{Change in Pot. Energy})$$

$$(wD)_{\text{ext}} = \frac{-Gmm_0}{r} - 0$$

$$\text{Grav Pot. at A} = V_A = \frac{(W.D)_{\text{ext}}}{m_0} = \frac{\Delta U}{m_0} \quad (\text{'Potential'})$$

$$V_A = \frac{-Gmm_0}{rm_0} = \frac{-Gm}{r} \quad (\text{scalar quantity})$$

Hence we can say

Pot. due to point mass at distance $r = \frac{-Gm}{r}$



Ek point mass ki wjah se r duri pr potential $\frac{-Gm}{r}$ aur gravitational field $\frac{Gm}{r^2}$ (towards the mass)

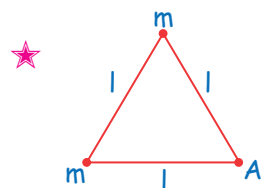
Hoti h ab ye mt kahna ki potential kidhar hoga hahahahaha gravitation potential Ek scalar quantity hai iski direction nhi hoti iska formula yad krlo aur formule ka minus mt bhlna



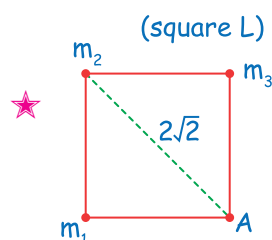
Find Grav. Potential at point 'A' in following case:



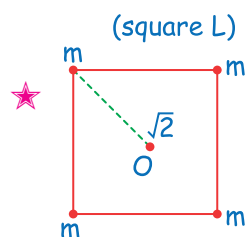
$$V_A = \frac{-Gm}{r}$$



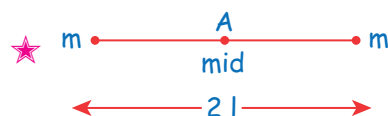
$$V_A = -\frac{Gm}{l} - \frac{Gm}{l} = -\frac{G2m}{l}$$



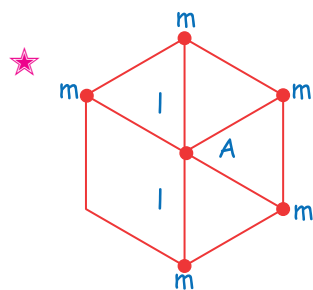
$$V_A = -\frac{Gm_1}{l} - \frac{Gm_2}{l\sqrt{2}} - \frac{Gm_3}{l}$$



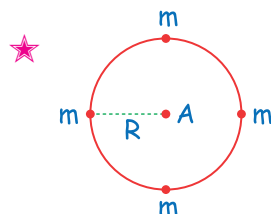
$$V_{at\ O} = -\frac{Gm}{l/\sqrt{2}} \times 4 = -\frac{G(4m)}{l/\sqrt{2}}$$



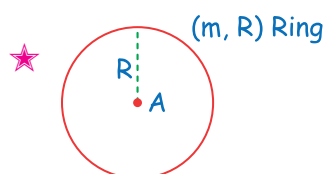
$$V_A = -\frac{Gm}{l} \times 2$$



$$V_A = -\frac{Gm}{l} \times 5$$



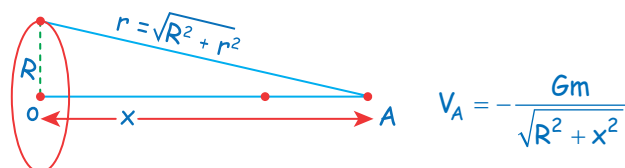
$$V_A = \frac{-Gm}{R} \times 4$$



RING (at centre)

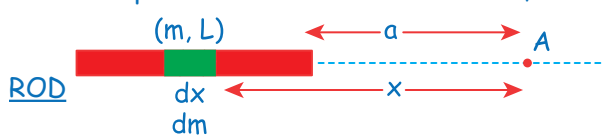
$$V_A = \frac{-Gm}{R}$$

Find potential at point A due to ring (at axial points)



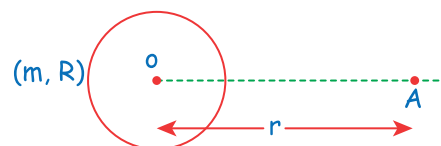
$$V_A = -\frac{Gm}{\sqrt{R^2 + x^2}}$$

Find potential at a due to thin rod m, L



$$\int dV = -\int \frac{G(dm)}{x} = \int_a^{a+L} -\frac{Gmdx}{Lx} = V = -\frac{Gm}{L} \ln\left(\frac{a+L}{a}\right)$$

Find potential due to hollow sphere



(1) $r > R$ (outside) (2) At surface

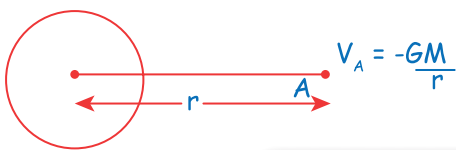
$$V_A = -\frac{Gm}{r}$$

$$V_A = \frac{-Gm}{R}$$

(3) Inside $r < R$

$$V_A = \frac{-Gm}{R}$$

- ★ Find potential due to solid sphere
(1) $r > R$ outside



- (2) At Surface

$$V = \frac{-Gm}{R}$$

- (3) Inside $r < R$

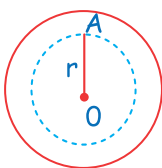
$$V_A = \frac{-Gm(3R^2 - r^2)}{2R^3}$$

so, At centre of earth

$$V_0 = \frac{-3Gm}{2R} \neq 0 \text{ (NON-ZERO)}$$

But (E_g) at centre = 0

Ye bhut hi important case hai bcz hanari earth ko v hm is chapter m solid sphere mante h



RELATION BETWEEN GRAVITATIONAL FIELD AND GRAVITATIONAL POTENTIAL

If $\vec{E}_g = E_x\hat{i} + E_y\hat{j} + E_z\hat{k}$

$$d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$$

$$\Delta V = -\int E_x dx + \int E_y dy + \int E_z dz = \int -\vec{E}_g \cdot d\vec{r}$$

or

$$E_x = -\frac{\partial V}{\partial x}; E_y = -\frac{\partial V}{\partial y}; E_z = -\frac{\partial V}{\partial z}$$

- Q. $\vec{E}_g = 2\hat{i} + 3\hat{j} + 4\hat{k}$. Find grav. Pot difference between (0, 2, 3) and (3, 6, 8)

Sol. $\Delta v = -\left[\int E_x dx + \int E_y dy + \int E_z dz\right]$

$$= -\left[\int_0^3 2dx + \int_2^6 3dy + \int_3^8 4dz\right]$$

$$= -[6 + 12 + 20] = -38 \text{ (SI unit)}$$

- Q. $v = x^2y$ find grav. field at (2, 3)?

Sol. $E_x = -\frac{\partial V}{\partial x} = -2xy$

$$E_y = -\frac{\partial V}{\partial y} = -(x^2)$$

$$\text{so } \vec{E}_g = -2xy\hat{i} - x^2\hat{j}$$

at (2, 3)

$$\vec{E}_g = -12\hat{i} - 4\hat{j}$$

- Q. If 2kg mass is placed at (2, 3). Find grav. force in previous question.

Sol. $\vec{F} = m\vec{E}_g = 2 \times (-12\hat{i} - 4\hat{j}) = -24\hat{i} - 8\hat{j}$

$$|\vec{F}| = \sqrt{24^2 + 8^2}$$

- Q. $V = x^2 + y^2 + z$ (let). Find grav. Field (E_g) at (1, 2, 3)

Sol. $E_x = -\frac{\partial V}{\partial x} = -[2x + 0 + 0] = -2x$

$$E_y = -\frac{\partial V}{\partial y} = -[0 + 2y + 0] = -2y^2$$

$$E_z = -\frac{\partial V}{\partial z} = -[0 + 0 + 1] = -1$$

$$\vec{E}_g = -2x\hat{i} - 2y\hat{j} - 1\hat{k}$$

at (1, 2, 3)

$$\vec{E}_g = -2\hat{i} - 4\hat{j} - 1\hat{k}$$

$$|\vec{E}_g| = \sqrt{2^2 + 4^2 + 1^2} = \sqrt{21}$$

- Q. $\vec{E} = 2x\hat{i} + 10y\hat{j} + 3z^2\hat{k}$. If Pot. at origin is 10J/kg find grav. Pot at (2, 3, 1).

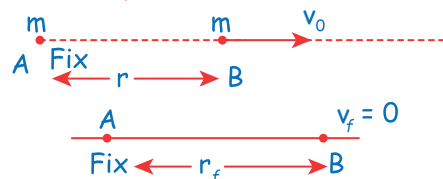
Sol. $\Delta V = V_f - V_i = -\left[\int_0^2 2xdx + \int_0^3 10ydy + \int_0^1 3z^2dz\right]$

$$V_f - 10 = -[4 + 5 \times 9 + 1] \quad [\text{To yahan } V_i = 0 \text{ mat le lena}]$$

$$V_f = -40$$

Question Based on M.E Conservation

- Q. Find where particle B will come to at rest



Sol. $K_i + U_i = K_f + U_f$

$$\left(\frac{1}{2}mv_0^2 - \frac{Gmm}{r}\right) = \left(0 - \frac{Gmm}{r_f}\right)$$

Now you can solve.

Q. A particle is projected from surface of earth with its velocity v_0 . Find max height attained by particle.

If ($v_0 < v_{\text{escape}}$)

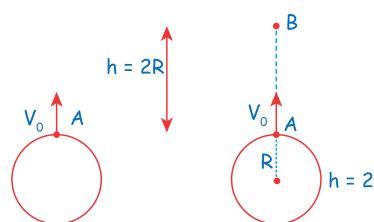
Sol. $K_A + U_A = K_B + U_B$

$$\frac{1}{2}mv_0^2 + mV_A = 0 + mV_B$$

$$\frac{1}{2}mv_0^2 - m\frac{Gm_e}{R} = -\frac{mGm}{(h+R)}$$

Now solve and get according to data

Q. With what minimum velocity a particle must be thrown vertically upward it reaches at a height equal to diameter of earth from surface B



Sol. $K_A + U_A = K_B + U_B$

$$\frac{1}{2}mv_0^2 - m\left(\frac{Gm}{R}\right) = 0 + m\left(\frac{-Gm_e}{3R}\right)$$

$$\Rightarrow \frac{1}{2}mv_0^2 = \frac{2}{3}\frac{Gm_em}{R}$$

$$V_0 = \sqrt{\frac{4}{3}g_0R} \quad \left(\because \vec{E}_g = \frac{-Gm}{R^3}\vec{x}\right)$$

Q. With what velocity particle will heat the ground

Sol. $K_i + U_i = K_f + U_f$

$$0 + m\left(-\frac{Gm}{2R}\right) = \frac{1}{2}mv_f^2 + m\left(-\frac{Gm_e}{R}\right)$$

$$V_f = \sqrt{\frac{Gm_e}{R}}$$

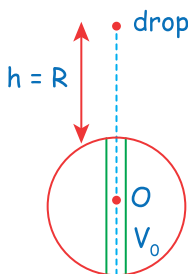
Q. Find speed of particle when reaches at centre of earth

Sol. $V_{\text{centre}} = V_{\text{max}}$

$$K_i + U_i = K_f + U_f$$

$$0 + mV_i = \frac{1}{2}mv_0^2 + m(V_{\text{centre}})$$

$$-m\left(\frac{Gm}{2R}\right) = \frac{1}{2}mv_0^2 + m\left(-\frac{3}{2}\frac{Gm}{R}\right)$$



$$V_0 = \sqrt{\frac{2Gm}{R}}$$

SHM X

$$F = -Kx$$

Oscillation ✓

Q. Find time period of oscillation if particle displace from Center of earth?

Sol. Solid sphere (Inside)

$$E_g = \frac{Gmr}{R^3}$$

$$\vec{F} = m\vec{E}_g$$

$$\vec{F} = \frac{-mGm\vec{x}}{R^3} \quad \left(\because \vec{E}_g = \frac{-Gm}{R^3}\vec{x}\right)$$

$\therefore \vec{F} = -k\vec{x}$ so motion is an SHM

$$\text{Time period, } T = 2\pi\sqrt{\frac{m}{k}}$$

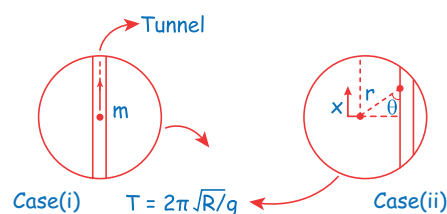
$$= 2\pi\sqrt{\frac{mR^3}{Gmm}} = 2\pi\sqrt{\frac{R^3}{Gm}}$$

Rearranging the expression.

$$T = 2\pi\sqrt{\frac{R}{\frac{Gm}{R^2}}} \quad \left(g = \frac{Gm}{R^2}\right)$$

$$T = 2\pi\sqrt{\frac{R}{g}}$$

Q. Repeat the above question if tunnel is not passing through centre of earth.



Sol. $F_{\text{net}} = F \sin\theta$

$$= mE_g \sin\theta = (mE_g) \sin\theta = m\frac{Gmr}{R^3} \sin\theta$$

$$F_{\text{net}} = \frac{Gmm}{R^3}(r \sin\theta)$$

Vector form $\vec{F}$:-

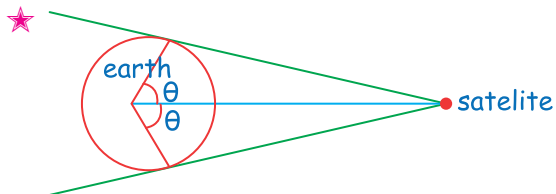
$$\vec{F}_{\text{net}} = \frac{-Gmm}{R^3}\vec{x} \quad (\text{in case ii also})$$

$$\text{so for both cases, } T = 2\pi\sqrt{\frac{R}{g}}$$



Kaam ka Dabba

	Grav. Field	Grav. potential
(1) Point Mass	$\frac{Gm}{r^2}$	$-\frac{Gm}{r}$
(2) Hollow Sphere	Outside $\rightarrow Gm/r^2$	$\rightarrow -Gm/r$
	Surface $\rightarrow GM/R^2$	$\rightarrow -GM/R$
	Inside $\rightarrow 0$	$\rightarrow -GM/R$
(3) Solid Sphere	Outside $\rightarrow Gm/r^2$	$\rightarrow -Gm/r$
	Surface $\rightarrow GM/R^2$	$\rightarrow -GM/R^2$
	Inside $\rightarrow \frac{Gmr}{R^3}$	$\rightarrow -\frac{Gm}{2R^3}(3R^2 - r^2)$



$$\text{Area Scan} = 2\pi R^2(1 - \cos\theta)$$

$$\text{Area Unscan} = 4\pi R^2 - 2\pi R^2(1 - \cos\theta)$$

$$\text{Inside } E_g = g = \frac{Gmr}{R^3}$$

$$\text{Surface } g = \frac{Gm}{R^2}$$

Outside

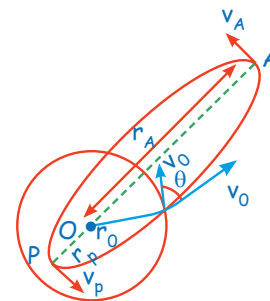
$$g = \frac{Gm}{r^2}$$

Ab apke concept ko build krne k liye kch question attach Kr rha hu solve them sincerely



Q. A small satellite of mass 'm' is revolving around earth in a circular orbit of radius r_0 with speed v_0 . At certain point of its orbit, the direction of motion of satellite is suddenly changed by angle $\theta = \cos^{-1}(3/5)$ by turning its velocity vector in the same plane of motion, such that speed remains constant. The satellite, consequently goes to elliptical orbit around earth. The ratio of speed at perigee to speed at apogee is:

Sol. Using conservation of angular momentum about O.



$$mV_{pI_p} = mV_{AI_A} = mV_0 I_0 \cos \theta$$

$$V_{AI_A} = V_{pI_p} = \frac{3V_0 r_0}{5}$$

Using conservation of energy

$$\frac{1}{2} mV_A^2 + \frac{-GMm}{r_A} = \frac{-GMm}{r_0} + \frac{1}{2} mV_0^2$$

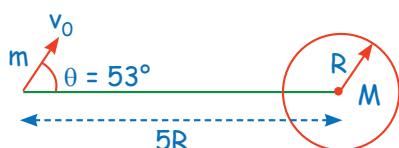
$$\Rightarrow \frac{9V_0^2 r_0^2}{50r_A^2} - \frac{V_0^2 r_0}{r_A} + \frac{V_0^2}{2} = 0 \left[\text{Let } \frac{r_0}{r_A} = x \right]$$

$$\Rightarrow 9x^2 - 50x + 25 = 0 \Rightarrow x = 5 \text{ or } (5/9)$$

$$\Rightarrow r_A = \frac{9r_0}{5}; r_p = \frac{r_0}{5}$$

$$\Rightarrow \frac{V_p}{V_A} = \frac{r_A}{r_p} = 9$$

- Q.** A spaceship is sent to investigate a planet of mass M and radius R . While hanging motionless in space at a distance $5R$ from the centre of the planet, the spaceship fires an instrument package with speed v_0 as shown in the figure. The package has mass m , which is much smaller than the mass of the spaceship. The package just grazes the surface of the planet and at that moment its speed is v . $v \cdot v_0 = \sqrt{\frac{xGM}{75R}}$. Fill x in OMR sheet.



Sol. $L_i = mv_0 \sin \theta \times 5R = mv \times R$

$$v = 5v_0 \sin \theta$$

$$\frac{1}{2} mv_0^2 - \frac{GMm}{5R} = \frac{1}{2} mv^2 - \frac{GMm}{R}$$

$$\frac{v_0^2}{2} - \frac{GM}{5R} = \frac{25v_0^2 \sin^2 \theta}{2} - \frac{GM}{R}$$

$$\frac{v_0^2}{2} [25 \sin^2 \theta - 1] = \frac{4GM}{5R}$$

$$\sin \theta = \frac{1}{5} \sqrt{1 + \frac{8GM}{5v_0^2 R}}$$

$$\frac{v_0^2}{2} \times 15 = \frac{4GM}{5R}$$

$$v_0 = \sqrt{\frac{8GM}{75R}}$$

- Q.** A satellite of mass m is orbiting the earth in a circular orbit of radius r . It starts losing energy due to small air resistance at the rate of CJ/s . Then the time taken for the satellite to reach the earth is

Sol. Energy in the orbit $= -\frac{GMm}{2r}$

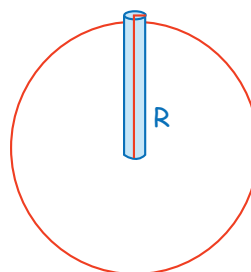
$$\text{Energy at the surface} = -\frac{GMm}{2R_e}$$

$$\text{Loss in energy} = \frac{GMm}{2} \left(\frac{1}{R_e} - \frac{1}{r} \right)$$

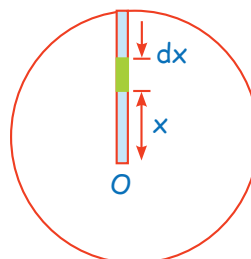
$$\text{time required} = \frac{\text{Loss in energy}}{C}$$

$$t = \frac{GMm}{2C} \left(\frac{1}{R_e} - \frac{1}{r} \right)$$

- Q.** Imagine a hole drilled along the radius of the earth. A uniform rod of length equal to the radius (R) of the earth is inserted into this hole. Find the distance of centre of gravity of the rod from the centre.



Sol. $x_{cg} = \frac{\int x dw}{\int dw}$



Where dw = weight of an infinitesimal element whose x co-ordinate is x .

Let l = mass per unit length of the rod.

Consider an element at a distance x from the centre

$$dw = (l dx) \left(g \frac{x}{R} \right)$$

[Acceleration due to gravity at a distance x from the centre of the earth is $= g \frac{x}{R}$]

$$\therefore x_{eg} = \frac{\frac{\lambda g}{R} \int_0^R x^2 dx}{\frac{\lambda g}{R} \int_0^R x dx} = \frac{\frac{R^3}{3}}{\frac{R^2}{2}} = \frac{2R}{3}$$

Q. At what distance above and below the surface of the earth a body will have same weight, (take radius of earth as R .)

Sol. Acceleration due to gravity at point p above the surface of earth

$$g_p = \frac{gR^2}{(R+h)^2}$$

Acceleration due to gravity at point q below the surface of earth

$$g_q = g \left(1 - \frac{h}{R} \right)$$

$$g_p = g_q$$

$$\frac{g}{\left(1 + \frac{h}{R} \right)^2} = g \left(1 - \frac{h}{R} \right)$$

$$\text{Take } \frac{h}{R} = x$$

$$\text{So } x^3 - x + x^2 = 0$$

$$x = \frac{\sqrt{5} - 1}{2}$$

$$h = \frac{R}{2} (\sqrt{5} - 1)$$

Q. The acceleration due to gravity on the surface of earth is g . If the diameter of earth reduces to half of its original value and mass remains constant, then acceleration due to gravity on the surface of earth would be:

Sol. As $g = \frac{GM}{R^2} \Rightarrow \frac{g_2}{g_1} = \frac{R_1^2}{R_2^2}$

$$\therefore g_2 = 4g_1 \left(R_2 = \frac{R_1}{2} \right)$$

Q. If R is the radius of the earth and the acceleration due to gravity on the surface of earth is $g = \pi^2 \text{ m/s}^2$, then the length of the second's pendulum at a height $h = 2R$ from the surface of earth will be:

Sol. $g' = \frac{GMe}{(3R)^2} = \frac{1}{9}g$

$$T = 2\pi \sqrt{\frac{\ell}{g'}}$$

For second pendulum,

$$T = 2 \text{ sec}$$

$$2 = 2\pi \sqrt{\frac{\ell}{g} \cdot 9}, \ell = \frac{1}{9} \text{ m}$$

Q. Assuming the earth to be a sphere of uniform mass density, a body weighed 300 N on the surface of earth. How much it would weigh at $R/4$ depth under surface of earth?

Sol. Since, $mg = 300\text{N}$

$$\therefore m = \frac{300}{g}$$

Acceleration due to gravity at depth of

$$g_d = g \left[1 - \frac{d}{R} \right]$$

$$g_d = g \left[1 - \frac{R}{4R} \right] \quad \left(\because d = \frac{R}{4} \right)$$

$$g_d = \frac{3g}{4}$$

$$\text{Weight at depth} = m \times \frac{3g}{4}$$

$$= \frac{3}{4} \times 300 = 225$$

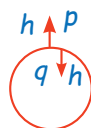
Q. If the acceleration due to gravity experienced by a point mass at a height h above the surface of earth is same as that of the acceleration due to gravity at a depth αh ($h \ll R_e$) from the earth surface. The value of α will be _____. (use $R_e = 6400 \text{ km}$)

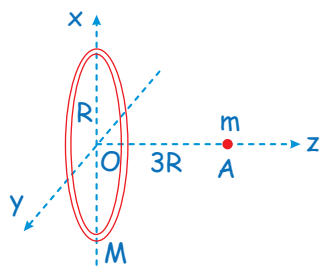
Sol. Given, $g_h = g_{\alpha h}, h \ll R$

$$\Rightarrow g \left(1 - \frac{2h}{R} \right) = g \left(1 - \frac{\alpha h}{R} \right)$$

$$\Rightarrow \frac{2h}{R} = \alpha \frac{h}{R} \Rightarrow \alpha = 2$$

Q. A circular ring of mass M and radius R is placed in xy plane with centre at origin. A particle of mass m is released from rest at a point $z = 3R$. Find the speed with which it will pass the centre of ring.





Sol. As shown in figure, first we find potential at A due to the ring.

$$V_A = -\frac{GM}{\sqrt{R^2 + (3R)^2}} = -\frac{GM}{\sqrt{10}R}$$

Now potential at centre of ring due to the ring is

$$V_0 = -\frac{GM}{R}$$

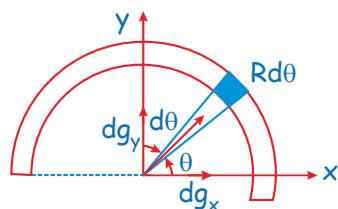
When the mass moves from A to O, work done on it due to gravitation forces is

$$W = m(V_A - V_0) = m\left(-\frac{GM}{\sqrt{10}R} + \frac{GM}{R}\right) = \frac{GMm}{R}\left(\frac{\sqrt{10}-1}{\sqrt{10}}\right)$$

This work done by gravitational force on mass must be equal to the increase in kinetic energy of the mass. Thus,

$$\frac{1}{2}mv^2 = \frac{\sqrt{10}-1}{\sqrt{10}} \frac{GMm}{R} \text{ or } v = \left[\frac{2(\sqrt{10}-1)GM}{\sqrt{10}R} \right]^{1/2}$$

Q. Find the field strength at the centre of a thin semicircular ring of radius R and mass M as shown in figure. Take the linear mass density as λ kg/m.



Sol. We choose an element that is an arc of length $ds = R d\theta$, its mass is $dm = \lambda R d\theta$. For every element at $+x$ there is an equivalent element at $-x$ whose contribution is symmetrically opposite. From this symmetry we can see that the field strength has no x-component at the centre.

The y-component of the field is

$$dg_y = dg \sin \theta = \frac{G dm \sin \theta}{R^2} = \frac{G \lambda R d\theta \sin \theta}{R^2} = \frac{G \lambda}{R} \sin \theta d\theta$$

The resultant field strength is

$$g_{\text{net}} = \int dg_y = \frac{G \lambda}{R} \int_0^\pi \sin \theta d\theta = \frac{2G \lambda}{R} \text{ (along + y direction)}$$

Q. A projectile is fired straight up from the surface of the earth with an initial speed $v_i = 9$ km/s. Neglecting air resistance, find the maximum height to which the projectile rises.

Sol. We will determine the maximum height by applying energy conservation law. We take earth's surface as initial point and maximum height as final point.

$$U_i + KE_i = U_f + KE_f$$

$$-\frac{GM_E m}{R_E} + \frac{1}{2}mv_i^2 = -\frac{GM_E m}{r} + 0$$

$$\frac{1}{2}v_i^2 = \frac{GM_E}{R_E} \left(1 - \frac{R_E}{r}\right) = g R_E \left(1 - \frac{R_E}{r}\right) \quad \dots (i)$$

$$\left[\because \frac{GM_E}{R_E^2} = g \right]$$

Equation (i) becomes

$$1 - \frac{R_E}{r} = \frac{v_i^2}{2g R_E}$$

$$\Rightarrow r = \frac{R_E}{1 - v_i^2 / 2g R_E}$$

$$\text{So, } \frac{v_i^2}{2g R_E} = \frac{(9000)^2}{2(9.81)(6.40 \times 10^6)} = 0.645$$

$$r = \frac{R_E}{1 - 0.645} = 2.81 R_E$$

$$\therefore h = r - R_E = 1.81 R_E$$